

Subject: Mobile Programming

Practical 3

Name: Ansari Muarif

Class: T.Y BSc. IT

Roll no: 3

Teacher Sign:

Q.] Write a program to create an Ionic application to accept the marks of five subjects. Calculate the total marks, percentage, result (Pass/Fail), and grade. Display the complete student result on another page using navigation.

Aim :-

To create an Ionic Angular application that accepts marks of five subjects, calculates total marks, percentage, result and grade, and displays the student result on another page using navigation.

Objective

- To create an Ionic Angular application.
- To accept marks using Ionic input components.
- To calculate total marks and percentage.
- To determine Pass/Fail result and grade.
- To understand page navigation in Ionic Angular.
- To display the complete result on another page.

```
import { Component } from '@angular/core';
```

```
import {
```

```
  IonApp,
```

```
  IonRouterOutlet,
```

```
  IonContent,
```

```
  IonInput,
```

```
  IonButton
```

```
} from '@ionic/angular/standalone';
```

```
import { Router } from '@angular/router';
```

```
@Component({
```

```
  selector: 'app-root',
```

```
  templateUrl: 'app.component.html',
```

```
  imports: [
```

```
    IonApp,
```

```
    IonRouterOutlet,
```

```
    IonContent,
```

```
    IonInput,
```

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```
IonButton

],

standalone: true,

})

export class AppComponent {

  constructor(private router: Router) {}

  marks = [0, 0, 0, 0, 0];

  showResult() {
    const total = this.marks.reduce((sum, mark) => sum + Number(mark), 0);
    const percentage = total / 5;

    let result = percentage >= 40 ? 'Pass' : 'Fail';
    let grade = '';

    if (percentage >= 75) {
      grade = 'A';
    } else if (percentage >= 60) {
      grade = 'B';
    } else if (percentage >= 50) {
      grade = 'C';
    } else if (percentage >= 40) {
      grade = 'D';
    } else {
      grade = 'F';
    }

    this.router.navigate(['/result'], {
      state: {
```

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```
marks: this.marks,

total: total,

percentage: percentage,

result: result,

grade: grade

}

});

}

}

<ion-app>

  <ion-content class="ion-padding">

    <h1>Student Marks</h1>

    <br>

    <ion-input label="Subject 1" type="number" [(ngModel)]="marks[0]"></ion-input>

    <ion-input label="Subject 2" type="number" [(ngModel)]="marks[1]"></ion-input>

    <ion-input label="Subject 3" type="number" [(ngModel)]="marks[2]"></ion-input>

    <ion-input label="Subject 4" type="number" [(ngModel)]="marks[3]"></ion-input>

    <ion-input label="Subject 5" type="number" [(ngModel)]="marks[4]"></ion-input>

    <br>

    <ion-button (click)="showResult()">

      Calculate Result

    </ion-button>

  </ion-content>

  <ion-router-outlet></ion-router-outlet>

</ion-app>

import { Component } from '@angular/core';
import { IonContent, IonButton } from '@ionic/angular/standalone';
import { Router } from '@angular/router';
```

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```
@Component({
  selector: 'app-result',
  templateUrl: './result.page.html',
  standalone: true,
  imports: [
    IonContent,
    IonButton
  ]
})
export class ResultPage {
  marks: number[] = [];
  total = 0;
  percentage = 0;
  result = "";
  grade = "";

  constructor(private router: Router) {
    const data = history.state;

    this.marks = data.marks;
    this.total = data.total;
    this.percentage = data.percentage;
    this.result = data.result;
    this.grade = data.grade;
  }

  goBack() {
    this.router.navigate(['/']);
  }
}

<ion-content class="ion-padding">
```

<h1>Student Result</h1>

<p>Subject 1: {{marks[0]}}</p>

<p>Subject 2: {{marks[1]}}</p>

<p>Subject 3: {{marks[2]}}</p>

<p>Subject 4: {{marks[3]}}</p>

<p>Subject 5: {{marks[4]}}</p>

<h2>Total Marks: {{total}}</h2>

<h2>Percentage: {{percentage}}%</h2>

<h2>Result: {{result}}</h2>

<h2>Grade: {{grade}}</h2>

<ion-button (click)="goBack()">

Back

</ion-button>

</ion-content>

import { Routes } from '@angular/router';

export const routes: Routes = [

{

path: '',

loadComponent: () =>

import('./app.component').then(m => m.AppComponent)

},

{

path: 'result',

loadComponent: () =>

import('./result.page').then(m => m.ResultPage)

}

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];

OUTPUT :-

Student Result

Subject 1	80
Subject 2	75
Subject 3	70
Subject 4	85
Subject 5	90
Total Marks	400
Percentage	80%
Result	Pass
Grade	A

BACK

Marks Result

Student Marks

Enter Marks

Subject 1	80	/ 100
Subject 2	75	/ 100
Subject 3	70	/ 100
Subject 4	85	/ 100
Subject 5	90	/ 100

CALCULATE RESULT

Marks Result